Edwin Arkel Rios Personal Page
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Architect of efficient deep learning systems, specializing in model compression and HW-SW co-design.

Education

National Yang Ming Chiao Tung University Hsinchu, Taiwan
PhD in Electrical Engineering and Computer Science, Cumulative GPA: 4.24/4.30 *July 2019 - Present*

National Cheng Kung University Tainan, Taiwan
Bachelor of Science in Energy Engineering, Cumulative GPA: 3.76/4.30 *Sept. 2015 - June 2019*

Experience

Tranxform AI, Generative AI R&D Intern *December 2025 - September 2026*

- Developed a data-free post-training quantization (PTQ) algorithm to quantize large audio language models (LALMs) to 1.58-bit precision, enabling up to 20x memory reduction (e.g., 28GB → ∼1.4GB for 7B models).
- Built a quantized automatic speech recognition (ASR) framework in Hugging Face transformers (32 quantization configs, 26+ models, 100+ datasets, 50+ languages) used in 5 papers (3 accepted).

NYCU PCS Lab & NTHU MIS Lab, Research Assistant *July 2019 - Present*

- Developed methods for knowledge distillation, pruning, and efficient training of vision models, leading to 8 first-author papers in international venues: ICASSP, ISCAS, ECCV Workshop, and CVPR Workshop.
- Led and mentored 20+ students, resulting in 10 publications and 3 awards at the school and national levels.

Selected Research Projects

Low-Bit Quantization of Large Audio Language Models *2025 - 2026*

- Benchmarked 18 ASR and audio language models across 25 languages at 3 to 8 bits: accuracy loss tracks model size, not per-language training data; distilled 1.58-bit decoders with 10x compression.

Input Pruning for Acceleration of Transformers via HW-SW Co-Design *2023 - Present*

- Designed a module to support pruning of up to 90% of inputs to reduce FLOPs while maintaining accuracy.
- Proposed a methodology to select the optimal settings for token reduction in an FPGA accelerator, increasing FPS by up to 20x and achieving 7x higher energy efficiency compared to SOTA accelerators.

Parameter-Efficient Fine-Tuning (PEFT) of Foundation Models *2023 - Present*

- Developed a lightweight adapter to adapt large models with limited resources, improving the accuracy by 5.1% over baselines while requiring 123x fewer trainable parameters.

Efficient Discriminative Feature Selection for Fine-Grained Image Recognition (FGIR) *2021 - 2024*

- Engineered a novel mechanism for selecting high importance regions for fine-grained classification, requiring up to 10^7 times fewer FLOPs while improving classification performance by 10.15%.

Skills

Languages: Spanish (native), English (fluent), Mandarin Chinese (proficient)

Programming Languages: Python, MATLAB, C++, Bash

Libraries: PyTorch, Hugging Face transformers, HQQ, bitsandbytes, scikit-learn, OpenCV, NumPy, pandas

Tools: Git, Docker, Linux, Weights & Biases, Vitis HLS for Xilinx FPGAs, Jupyter Notebooks, LaTeX, HTML

Awards and Certifications

Merit Paper Award, 39th IPPR Conference on CV, Graphics and Image Processing (CVGIP) *2026*

Excellent Paper Award, 38th IPPR Conference on CV, Graphics and Image Processing (CVGIP) *2025*

Associate Fellow (AFHEA) of AdvanceHE, UK Professional Standards Framework in Higher Education *2025*

NCTU Academic Achievement Award during R.O.C. Academic Semester 108-1 *2020*

NCKU Outstanding Student during R.O.C. Academic Year 105 and 106 *2018*

Selected Research Works

10 first-author papers (ICASSP, ISCAS, ECCV/CVPR/EMNLP/NeurIPS W); 1 ICLR talk; 6 in review.

How to Choose Your Teacher for Fine Grained Image Recognition *CVPR W 26*

Cross-Layer Cache Aggregation for Token Reduction in Ultra-Fine-Grained Image Recognition *ICASSP 25*